CRASH COURSE #2 · THURSDAY 22 OCTOBER 2026 · 17:00 → 19:00 · ON ZOOM

Vibe coding & AI-assisted development

Direct the agent. Check the result.

Where AI agents come from, how they change research, and how to work with one: quizzes and discussion, on real data.

FacMed Hack #1 · Digital Phenotyping of Mental Health · Fri 6 – Sat 7 November 2026 · Campus Biotech · in English

UNIVERSITÉ DE GENÈVE
FACULTY OF MEDICINE
Synapsy centre for neuroscience and mental health research

WYSS CENTER

EPFL

neuro @campusbiotech

HUG Hôpitaux Universitaires Genève

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SWISS CENTER FOR AFFECTIVE SCIENCES

HPE

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RESEARCH AND GRANTS OFFICE

Sparkboard

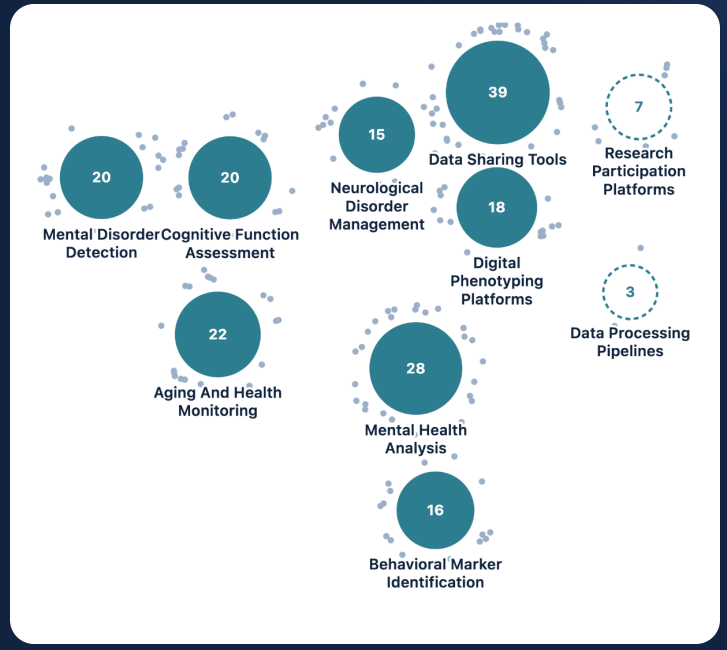
BEFORE ANYTHING ELSE · THANK YOU

You built the needs library

188

needs, against a goal of **100**

Almost twice the goal, from **30 contributors** at UNIGE, HUG, the Wyss Center, FCBG, Trieste and UNIL. Prizes for the best needs are announced at the finals.





Where the needs are: topics (left) · domains × methods (right) · facmed-hack.sparkboard.com/needs, 22 Oct

The road to 6 November

THU 8 OCT
Crash Course #1
Digital phenotyping & the Swiss data rules

TODAY
Crash Course #2
Vibe coding & AI-assisted development

23 → 28 OCT
Challenge voting
Rank your three favourite challenges and confirm your place

TUE 3 NOV · 17:00
Crash Course #3
Grant writing for pop-up projects, with the Research and Grants Office (RGO)

6 – 7 NOV
The hackathon
32 hours at Campus Biotech, in English



FROM HERE TO 6 NOVEMBER

From your needs to the challenges

1 · NEEDS

188 needs

Posted by researchers and clinicians; call closed 15 October.

2 · CLUSTERS

Grouped

By the technical problem a team would work on, across labs and institutions.

3 · SELECTION

Steering committee

14 October: 8 candidate challenges, aiming for about 7.

4 · SHAPING

Challenge sheets

A postdoc runner, data ready on day 1, compute and privacy routes.

5 · YOU

Vote and teams

23 → 28 Oct: rank three challenges; teams are formed from your rankings.

EIGHT CHALLENGES TAKING SHAPE · DRAFT OF 14 OCTOBER · FINAL LIST ONLINE ON 23 OCTOBER

- **1** · Between-visit deterioration
- **2** · Physiology you can trust
- **3** · Behaviour annotation from video
- **4** · Clinical speech and language
- **5** · Markerless movement, gaze, pupil
- **6** · Digital markers ↔ the brain
- **7** · Cross-species behaviour and touch
- **8** · Interactive interventions

Tonight on Zoom

- **Camera on** if you can; **microphone off** unless you speak.
- **Questions in the chat**, any time.
- **Live quiz** on your phone or laptop: next slide.
- **Quizzes, discussion**, and **breakout rooms** at 18:43: unmute and argue.
- **Course folder and slides**: link in the chat.
- **We end at 19:00** sharp.



LIVE QUIZ · FIRST QUESTION · JOIN NOW

What have you already used AI for in your research? *Tick all that apply.*

- **A** Finding and summarising papers
- **B** Analysis code
- **C** Modelling
- **D** Writing and editing
- **E** Learning or explaining code and methods
- **F** Building small tools
- **G** Nothing yet

VOTE LIVE · PIN 683758

facmed-quiz.pages.dev

> *"What would you do if you had all the models of the brain you wanted?"*
>
> Grace Lindsay, NYU, responding to this week's pre-read (DePasquale, *The Transmitter*, June 2026)

OPENING QUESTION · KEEP IT IN MIND: WE COME BACK TO IT AT 18:53

Where do you imagine vibe coding and agentic coding will change science?

Name **one place, and what changes there**. Your own field or any other: a method, an instrument, a job in the lab, a stage of publishing. Be specific.

TONIGHT · THREE OBJECTIVES

By 19:00 you should…

1 · IMAGINE

See your place in agentic science

Where these tools come from, what they already do in research, and what becomes yours to do: your contribution, your career, your purpose.

2 · UNDERSTAND

Know how vibe coding works

Think fast and slow. Try, fail and retry cheaply. Git as the safety net, run by the agent for you. Working as a team, with agents in the loop.

3 · PREPARE

Know how it runs at the hack

Whether you already have the tools or not, both paths are covered: your own setup, or the HPE / NVIDIA environment, open to every team to build and test on.

Built on the course *Vibe Coding for Scientists* (github.com/wazaahhh/vibe-coding-crash-course, CC BY-NC 4.0). Pre-reads: DePasquale (2026) and Schuck (2026), both in *The Transmitter*.

25 min

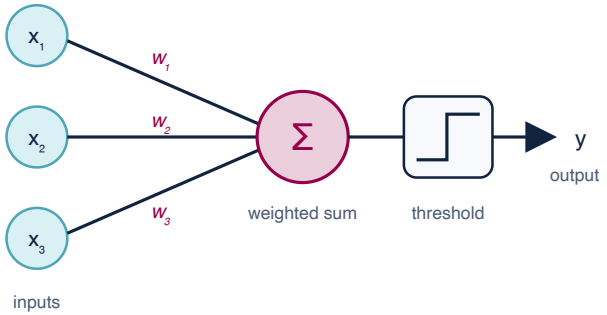
17:10 · PART 1 OF 7 · OBJECTIVE 1 · IMAGINE

Where this comes from

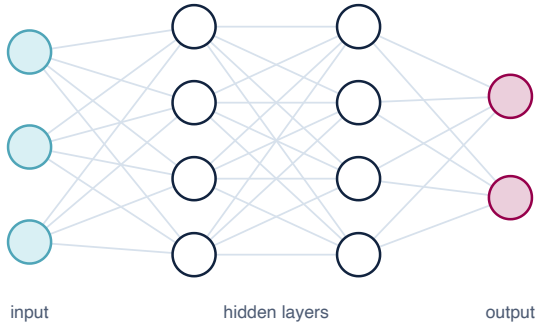
From an artificial neuron in 1943 to an agent that edits your analysis code. What these tools are, what scientists do with them, and where they fail.

It started in your field

YEAR	WHAT HAPPENED
1943	**McCulloch & Pitts** model a neuron as a logic unit: the first artificial neuron
1958	**Rosenblatt's perceptron** learns its weights from examples
1982	**Hopfield networks**: memories as stable states of a network
1986	**Backpropagation** (Rumelhart, Hinton & Williams): how multi-layer networks learn
2012	**Deep learning** wins image recognition (AlexNet): GPUs + data + depth
2024	**Nobel Prize in Physics** to Hopfield and Hinton



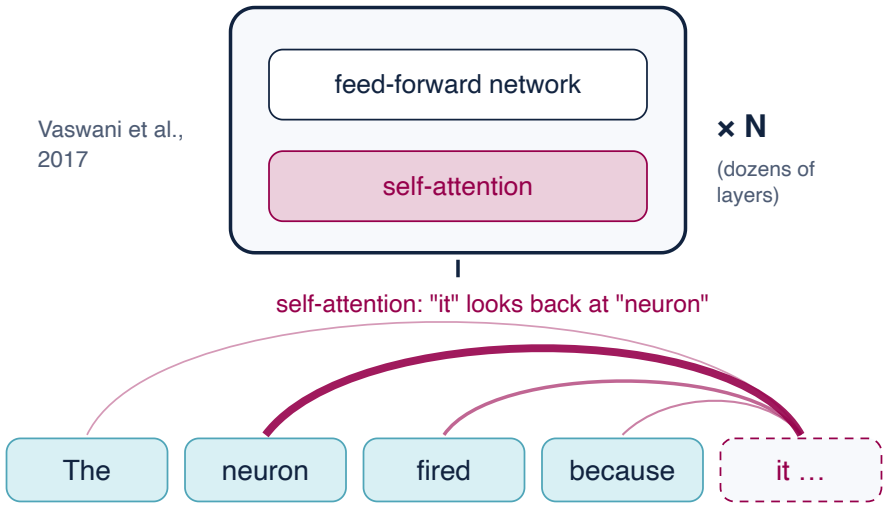
Perceptron: weighted inputs → sum → threshold



Multi-layer network, trained by backpropagation

From predicting the next word to editing your code

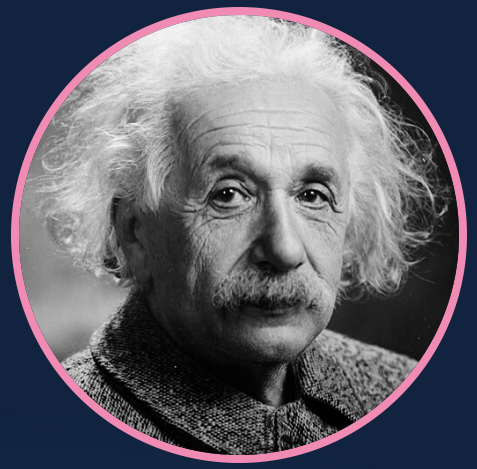
YEAR	WHAT HAPPENED
2017	**The Transformer**: Vaswani et al., "Attention is all you need", *NeurIPS* · arXiv:1706.03762
2020	**GPT-3**: 175 billion parameters; scale brings surprising abilities
2021	**GitHub Copilot** autocompletes code in the editor
Nov 2022	**ChatGPT**: language models reach everyone
2023–24	Claude, GPT-4, open-weight models; **"reasoning" models**
2025	**Coding agents** work inside your project; Karpathy names **"vibe coding"**
2026	Agents in research labs (this week's two pre-reads)



A Transformer layer: each word attends to the others, then a feed-forward step; stacked dozens of times

Why "attention" matters to you

To write each token, the model weighs every token in its context. Your question, codebook and constraints are what it attends to: **your attention steers its attention.**



"To raise new questions, new possibilities, to regard old problems from a new angle, requires creative imagination and marks real advance in science."

Albert Einstein & Leopold Infeld · *The Evolution of Physics*, 1938

Portrait: Oren Jack Turner, 1947 · public domain, via Wikimedia Commons

EXERCISE · 3 MINUTES · LIVE QUIZ

Same data, your question

The phone study (fictional data, real files)

150 students, 12 weeks, one phone app, consent on file.
Every day, passively: steps · how far they roam (GPS radius) · screen-on minutes · phone-estimated sleep · number of calls and messages (counts only, no content).
Every evening: one-tap mood, 1 to 7.
Weeks 0, 6, 12: PHQ-9 depression questionnaire.
Also: age, gender, faculty; exam weeks flagged.

What would you find worth exploring in this data? A few words, in the live quiz.

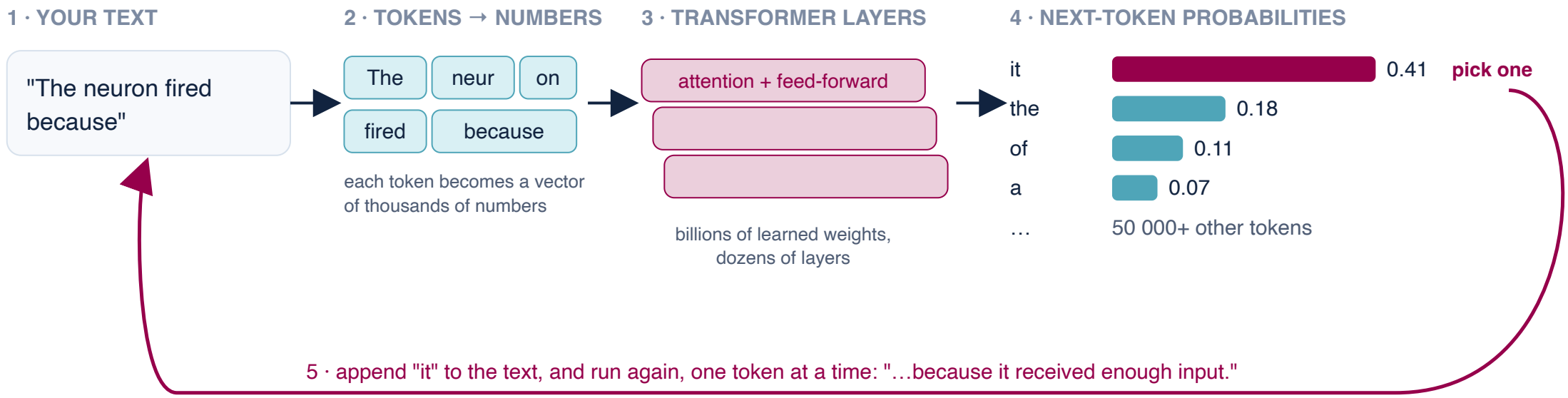
LIVE QUIZ

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Then we look at the word cloud: how many different questions did one dataset raise? The files are in the course folder, `phone-study/`.

How a large language model (LLM) writes, one token at a time



Illustrative numbers. Training set the weights by predicting the next token on huge text corpora; tuning taught it to follow instructions.

Fluent ≠ true

Most plausible, not proven: it can invent a citation.

Cut-off

Knows nothing after its training data.

Context

Only "sees" what fits in its window.

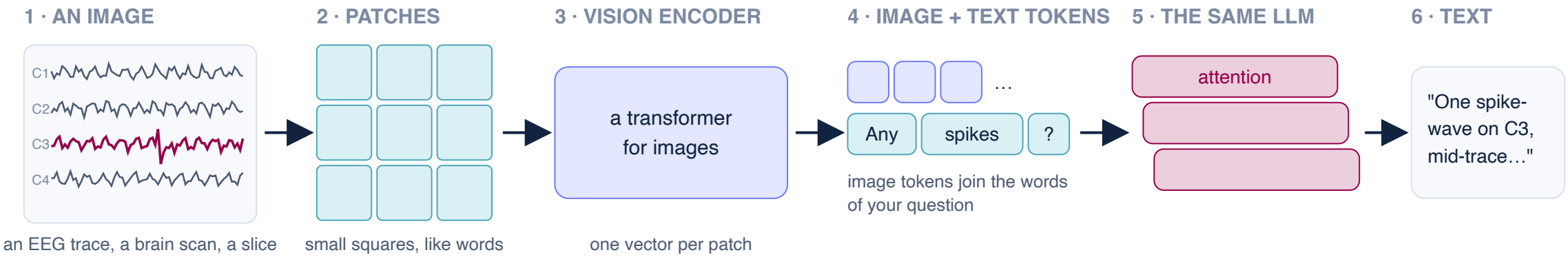
Not deterministic

Same question, maybe another answer.

Temperature

The dial on step 4. Low: safe, repeatable. High: bolder, more creative, more errors.

Vision-language models (VLM) in a nutshell



Every word can attend to every patch: the model "looks" where your question points it.

Schematic. Trained on huge sets of image–caption pairs, then tuned to answer questions about images.

What it adds

Reads EEG traces, brain scans, histology, figures and screenshots of errors, next to your text.

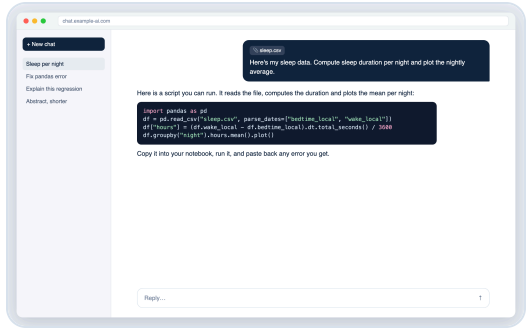
Why agents use it

An agent can **look at the figure it just made** and fix the axis, the colours, the missing panel.

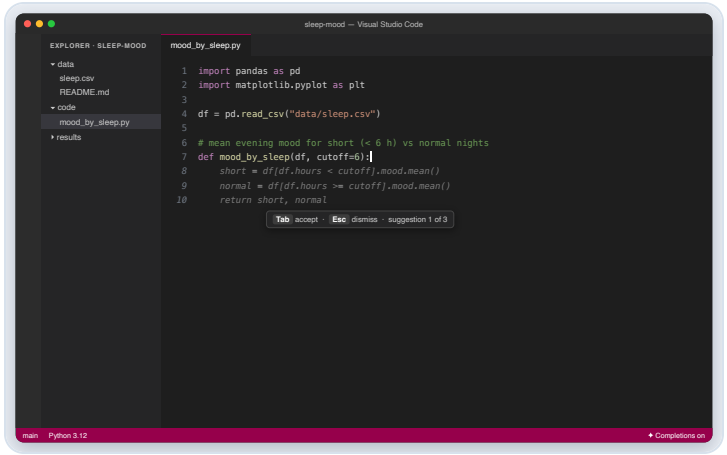
Where it fails

Counting, exact values, small print: it describes what it *expects* to see. And a face is personal data.

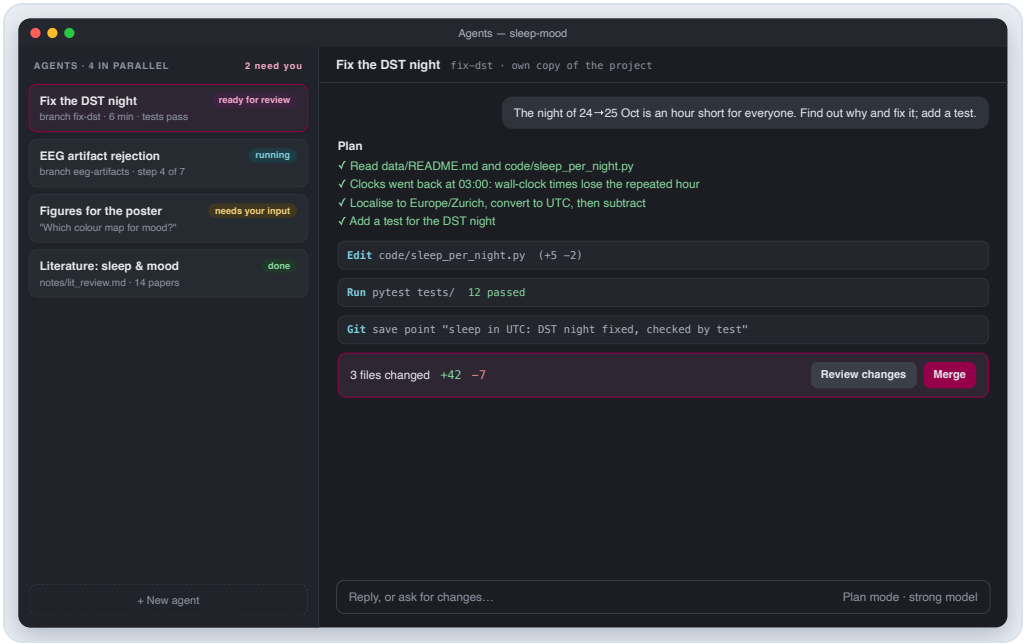
From chatbot, to code completion, to agentic coding



1 · Chatbot (browser, app): you paste code, data and errors; **you** carry the context and copy the code back.



2 · Code completion (VS Code): it suggests the next lines as you type; Tab accepts. **You** still drive every line.



3 · Agentic coding (e.g. Claude Code, Codex): several agents in parallel, each reads, plans, edits, runs and saves on its own branch; **you** review and merge.

> An agent is an LLM **in a loop with tools**: goal → plan → act → look at the result → adjust → report.

Software development is where agentic AI is used most

Why coding first

- Code is text, and there is a vast public corpus of it.
- Code can be **run**: errors and tests give immediate feedback.
- The result can be **checked**, by the machine and by you.

37%

of conversations with one major assistant were software and maths tasks, by far the first use (Anthropic Economic Index, 2025); still first in 2026, and moving from chat to **autonomous agent workflows**.

"Vibe coding" (Andrej Karpathy, 2025)

- You describe **what you want** in plain language; the model writes the code.
- Tool-agnostic: Claude Code, Codex, Copilot, Cline, or paste-into-chat.

> The honest version for scientists: the vibe is in **how you produce** code, never in whether it's correct.

A hackathon is a sprint with a deadline

CREATIVITY

Think out of the box

People from different labs, clinics and fields on one fresh problem. Every background brings another angle on the same data.

LEARNING BY DOING

Build to understand

You learn a method, a dataset or a tool by making something with it, next to someone who knows the part you don't.

A PINCH OF CHALLENGE

Ship by the deadline

At the end, each team shows a working **prototype or proof of concept**, not a plan. The clock forces the choices.

Time-bounded sprint = creativity × learning by doing × a deadline. FacMed Hack: 32 hours, Friday 6 → Saturday 7 November.

Vibe coding changed hackathons. It did not replace the team.



2019: a team of four declares five technologies · **2025**: one person with an agent, seven, by Sunday

Maillart et al., work in progress: every Devpost hackathon, 13,626 events, ~400,000 projects, 2014–2026.

What changed, sharply

- Generative AI in **53%** of projects, and growing by about **3 points every quarter**.
- Solo vs team of four: before AI, the team built with almost **twice as many technologies** (5.3 vs 2.9); with agentic AI, **the same** (7.4 vs 7.3).

What did not

- Going solo still lowers your **chance of winning by 22 points**, exactly as before AI.
- Teammates build something new together **62×**, and team up again **134×** more often than same-event strangers.

> AI made each person broader. **The team still wins, and the team outlasts the project.**

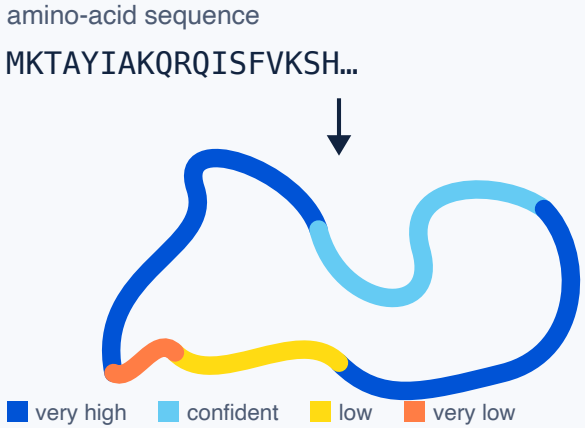
Six everyday uses

EVERYDAY USE	WHAT IT DOES FOR YOU
Literature	find, summarise, compare papers
Analysis code	cleaning, statistics, figures, pipelines
Modelling	from a word model to a fitted model in hours
Writing	editing, restructuring, translating
Learning	explaining unfamiliar code or methods
Tools	small apps, dashboards, lab utilities

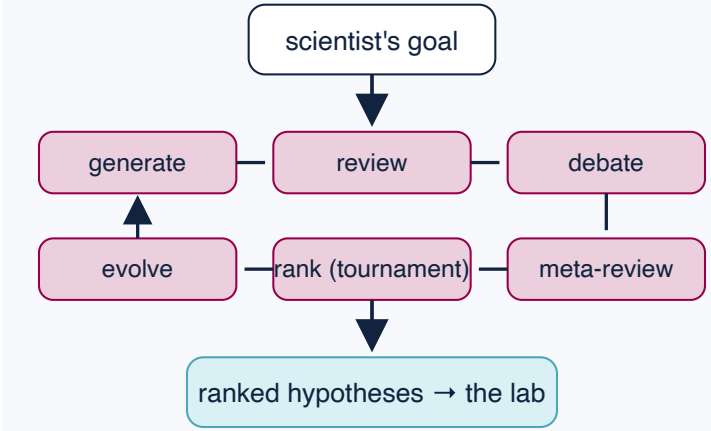
> Compare with the room's answers to the opening quiz: which uses are we missing?

AT THE FRONTIER

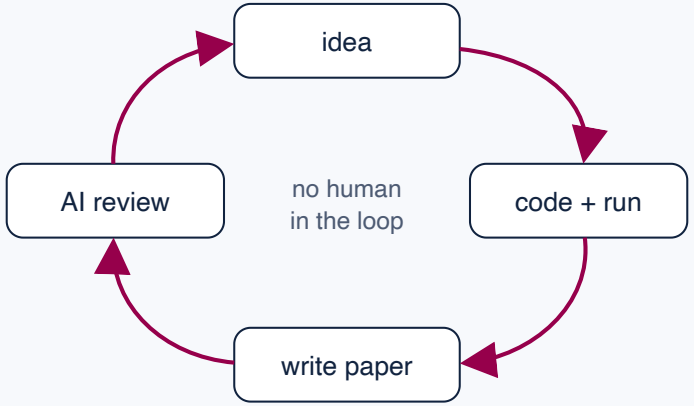
Beyond the chat window



AlphaFold predicts protein structures from sequence · Nobel Prize in Chemistry 2024



AI "co-scientist": agents generate, debate and rank hypotheses · Google, 2025



The AI Scientist drafts whole studies end to end, with mixed quality · Sakana AI, 2024

IN NEUROSCIENCE · LUO ET AL., *NATURE HUMAN BEHAVIOUR*, 2024

Can an LLM predict a neuroscience result better than you?

- **BrainBench**: 200 real *J. Neurosci.* abstracts, each paired with a version whose results were altered. Which one is real?
- LLMs beat **171 neuroscience experts**, in every subfield.
- A small model further trained on neuroscience papers (**BrainGPT**) did ~3 points better still.

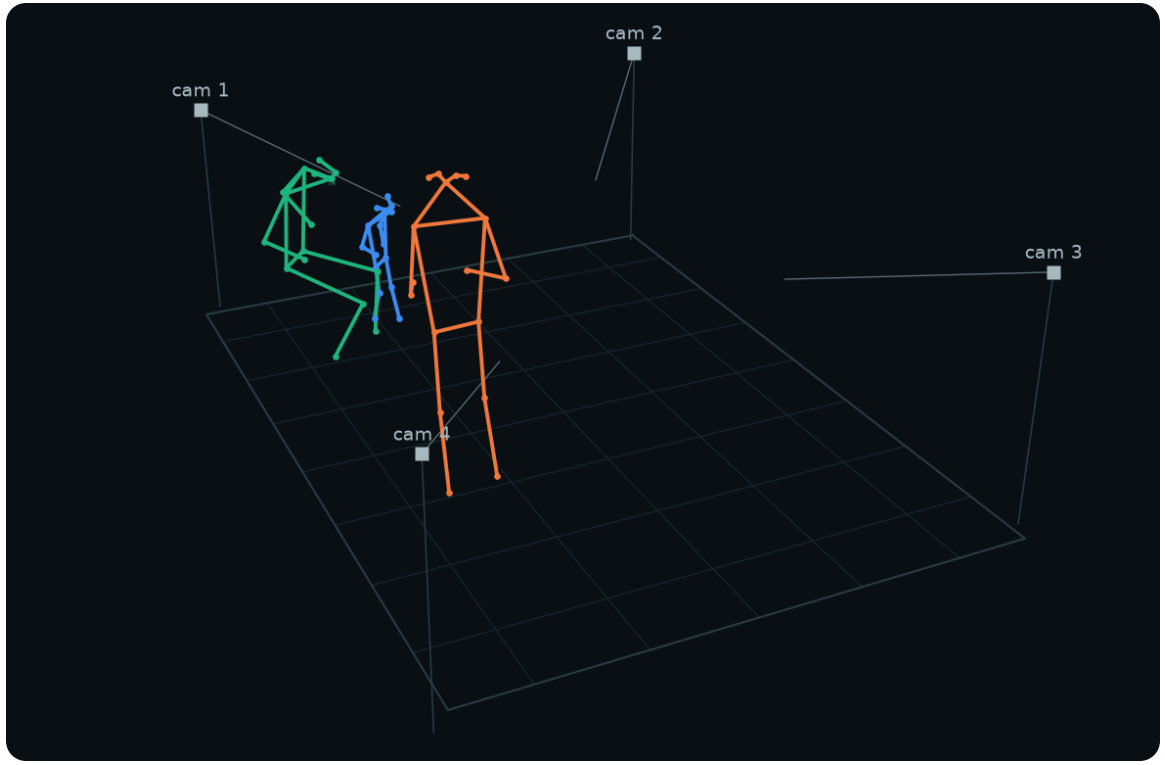
81%
LLMs

63%
human experts

doi.org/10.1038/s41562-024-02046-9 · Predicting what was published is not discovering: models learn the literature's patterns, biases included.

3D autism diagnosis scene reconstruction: unstuck after years

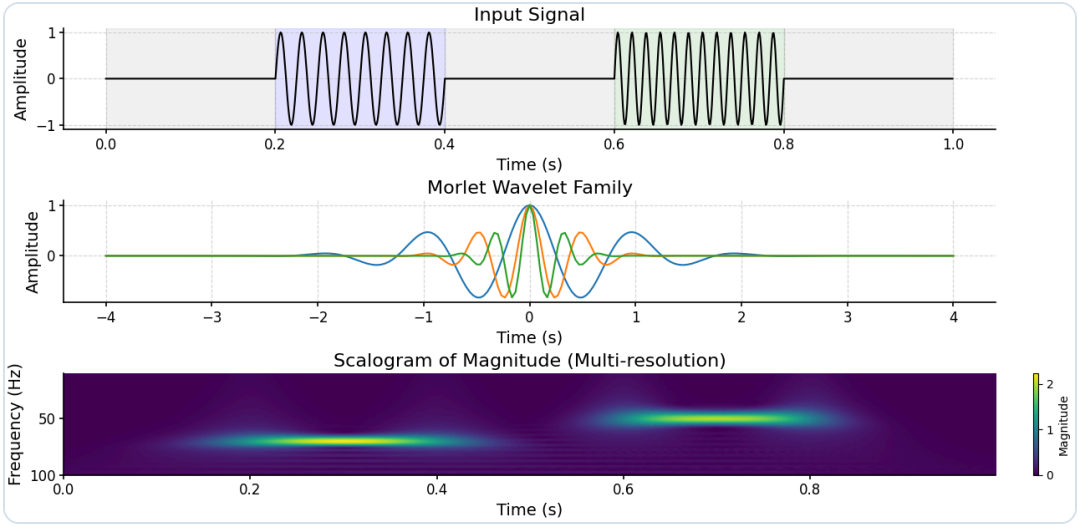
- **Stuck for years**: our own lack of technical skills.
- **With an agent**: three state-of-the-art methods pulled in: camera **synchronisation**, camera **calibration**, **triangulation** to 3-D.
- **Outcome**: hundreds of hours of 4K video, in storage since 2020, can finally be processed.



Four calibrated corner cameras → one metric 3-D scene · child (blue), clinician (orange), parent (green)

Detecting hand-flapping: many approaches tested, fast

- **Goal**: detect hand-flapping automatically in pose estimation of autistic children.
- **With an agent**: our approach rebuilt fast, then many others tested, ours and published ones.
- **Outcome**: no satisfactory result yet, but new research directions in sight. **There is a lot of learning in failing.**



One of the approaches, on a synthetic signal: oscillation → wavelets → scalogram

POLL 1 OF 6 · WHAT HOLDS YOU BACK

What holds you back most from using coding agents in your research? *Pick up to two.*

- **A** Not sure I can trust the results
- **B** Patient data and confidentiality rules
- **C** No time to learn a new way of working
- **D** Access, cost or institutional rules
- **E** Nothing: I already use them freely

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THE OPPORTUNITIES

Code was friction between a question and an answer

The plumbing stops being the bottleneck

Hours of searching forums and documentation become minutes.

Exploration gets cheap

Five model specifications, three plots, two normalisations, in the time of one.

The barrier drops for the whole lab

The student who "can't code" can build a working pipeline; experimentalists can model.

Scripts over clicks

An analysis written as code can be re-run, reviewed and shared: better for reproducibility.

> ⚠️ It is also trivially easy to produce **plausible · beautiful · wrong**.

Models at the speed of thought: four directions

1 · Experimentalists model

Fit a model to your own data without waiting months for a modelling collaborator.

2 · Theorists iterate

Try, test and reject new models in hours, at last at the speed theory promised.

3 · Search model space

AI-guided search (AlphaEvolve-style) explores many candidate models at scale.

4 · Maths opens up

Mathematically heavy models become reachable for people stronger at code than at the blackboard.

> *"Major new discoveries are likely to come from humans generating insights, just as they have always done..."* Larry Abbott, Columbia, in his response

"Prolific but shallow"?

> *"The field risks becoming prolific but shallow—generating models faster than we can generate insights."*
>
> Brian DePasquale

- **Thinking by struggling**: turning an idea into equations and code is *"a form of disciplined thinking"*, above all for trainees.
- **Hidden errors**: code that seems to test your idea but does not *"crop up very regularly in practice"* (Dan Goodman, Imperial).
- **Invented concepts**: only expertise recognises what *"does not actually exist"* (Julijana Gjorgjieva, TUM).

THE PITFALLS

Strongest on syntax, weakest on truth

- **Trust, but skim**: plotting, formatting, boilerplate, reading files, explaining code.
- **Slow down**: units, time zones, the statistical test, your specific method, citations.
- **It fails quietly**: a clean run with a wrong number, never a crash. `dropna()` drops 40% of rows; leakage gives a 0.98 that evaporates.
- **Beyond the code**: patient data pasted into cloud tools is a breach, whatever the answer.

WHAT THE SCIENCE SAYS

The danger is a confident answer that switches your checking off

- **Automation bias & complacency**: we over-trust automated output and stop checking it *(Parasuraman & Manzey, 2010)*.
- **Cognitive offloading**: the more knowledge workers trust generative AI, the less critical thinking they report *(Lee et al., CHI 2025)*.
- **Illusion of explanatory depth**: fluent output makes you *feel* you understand what you couldn't rebuild *(Rozenblit & Keil, 2002)*.
- **"Cognitive debt"**: heavy LLM use while writing went with weaker memory of one's own text and more homogeneous prose *(Kosmyna et al., 2025, preprint)*.
- **Generation effect**: you understand and retain what *you* produce far better than what you only read *(Slamecka & Graf, 1978)*.

> Offload the typing. **Never offload the understanding** your name depends on.

POLL 2 OF 6 · A REALITY CHECK

In a 2025 randomised trial, experienced open-source developers worked on their own projects with and without AI tools. With AI, they were…

- **A** about 40% faster
- **B** about 20% faster
- **C** no faster and no slower
- **D** about 19% slower

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POLL 2 · THE ANSWER

D: about 19% slower, while feeling 20% faster

+19%
time taken with AI tools, measured

−20%
time the same developers *believed* AI had saved them, afterwards

- 16 experienced developers, 246 real tasks on their own projects, early-2025 tools.
- METR's own caveat: experts in large codebases they knew by heart; it may not hold for small new projects, closer to a hackathon.
- **The lesson is the gap**: your feeling of speed is a poor measurement. That is thinking fast and slow, in one number.

METR (Model Evaluation & Threat Research), "Measuring the impact of early-2025 AI on experienced open-source developer productivity", July 2025.

THE THESIS WE'LL REPEAT

You move from author of code to director of code.

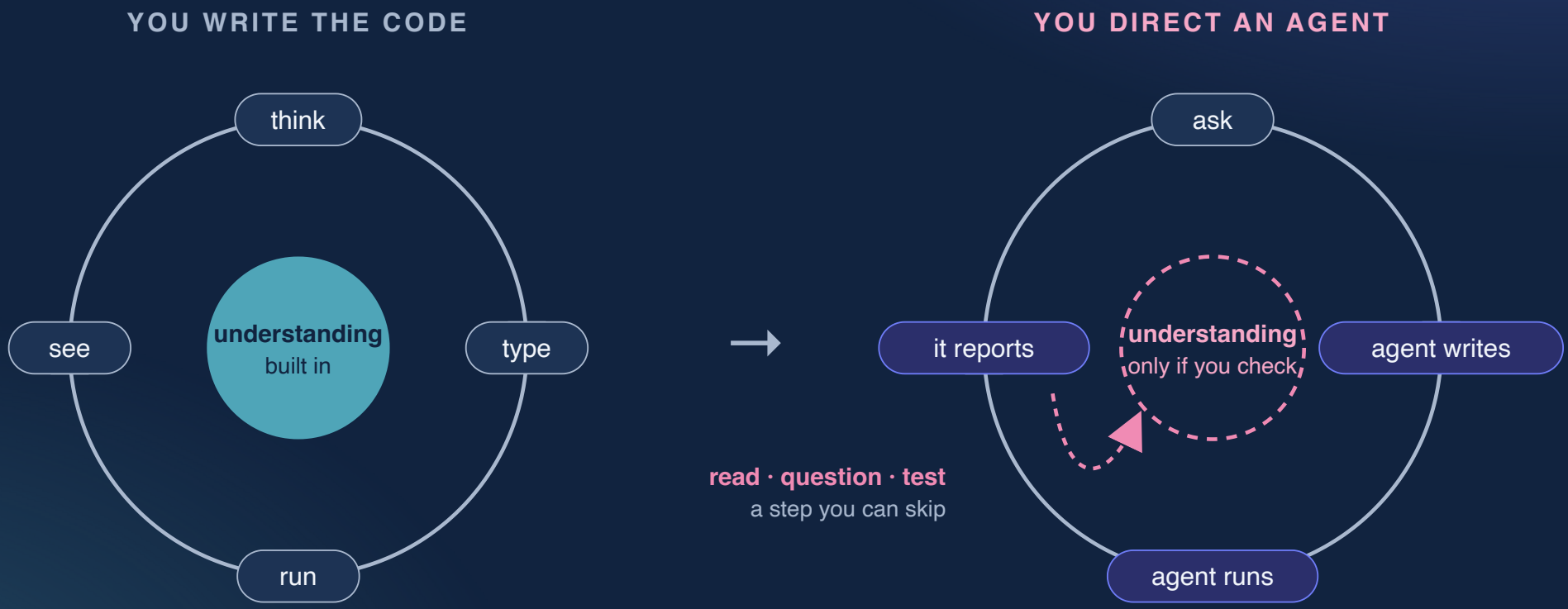
Your job becomes **intent · judgment · verification**. The syntax is the agent's.

Next: how to do that job well.

15 min

How vibe coding works

When you write code yourself, understanding comes (almost) for free. When you direct an agent, you have to *choose* to engage it. Four disciplines make the new loop safe.



Diverge first, then think critically

1 · Divergence

Science trains **convergence**: narrow to the one right method. Generation is now nearly free, so **diverge first**: *"Give me three approaches, with trade-offs."* Curate rather than produce; converge later, with more on the table.

2 · Critical thinking

The model is confident, fluent and occasionally fabricating, in flawless prose. Treat every output as a pull request from a brilliant, overconfident intern. Ask **"How would I know if this is wrong?"** *before* trusting the pretty plot.

> Trust the vibe to **write** it; never trust the vibe that it's **right**.

Think fast and slow; try and fail

3 · Thinking fast / slow



Daniel Kahneman



Amos Tversky

Heuristics and biases, from the 1970s: the work behind *Thinking, Fast and Slow* (2011).

The AI is your **System 1**: fast, fluent, sometimes wrong. **You** supply System 2: slow, deliberate, checking. *"System 2 is in charge of doubting, but System 2 is often lazy"* (Kahneman).

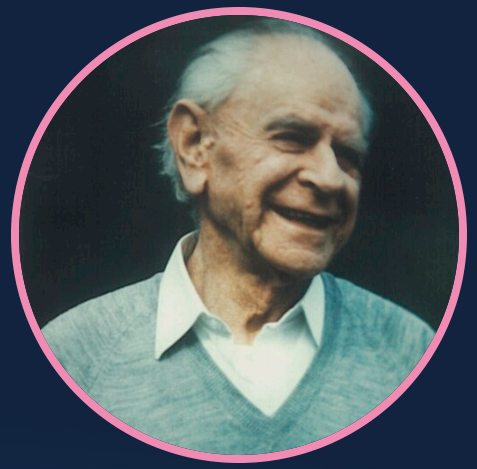
4 · Try and fail

Not a Silicon Valley motto: **the method of science**. A hypothesis is a guess you try to break; an experiment that fails tells you something (Popper, next slide).

What changes: a failed attempt now costs **minutes, not weeks**. More guesses, tested harder, as long as you can always go back (Part 3: git).

> The more the result matters, the more slowly you must read it.

Photo of D. Kahneman: NRKbeta.no, CC BY-SA, via Wikimedia Commons (cropped).



"The method of science is the method of bold conjectures and ingenious and severe attempts to refute them."

Karl Popper · *Objective Knowledge*, 1972

Photo of Karl Popper, 1987 · CC BY-SA 4.0, via Wikimedia Commons (cropped)

PART 2 OF 7 · FROM PRINCIPLES TO PRACTICE

Now, let's get concrete

How to set up a vibe-coding project, how to ask, and which dials to turn: the moves you will use at the hack, and in your lab.

1 **One folder, one rule file**

2 **How to ask**

3 **1 task = 1 agent**

4 **Three dials**

5 **A worked rhythm**

One project folder, one rule file

- Put **code · figures · notes · the paper** in one project folder: the agent sees the whole picture.
- Raw or private **data stays out of what gets shared** (Part 3), and **identifiable participant data stays out of the agent's reach**.
- A rule file at the top, **`AGENTS.md`** (or `CLAUDE.md` for Claude Code), tells the agent your house rules, every time.

> Chat answers a question. A project folder lets the agent **work on your actual project**.

```
my-project/
├── AGENTS.md       ← house rules for the agent
├── NOTES.md        ← what we verified / trusted
├── code/           ← analysis, notebooks
├── results/        ← figures, tables
├── data/           ← de-identified only; read, never modify
└── participants/   ← identifiable: OUT OF REACH
```

```
# Rules
- Python, pandas. Figures go to results/.
- Never modify files in data/.
- NEVER open, read, list or search participants/,
  or any copy of it, anywhere.
- Never open video (.mp4 .mov), audio (.wav .m4a),
  raw recordings (.edf .bdf), consent forms (.pdf),
  or files whose name contains "id", "name" or "contact".
- If a task needs them: stop and ask me.
- Plan first and wait for my OK.
```

How to ask: direct it like a capable new student

- **Give context, not just the task**: the data, the goal, the constraints, what you've tried.
- **Ask for a plan before code**: *"outline your approach; I'll approve, then you write it."*
- **Constrain it**: language, libraries, *"match the style in this file"*, *"no new dependencies"*.
- **Make it show its work**: *"explain your reasoning; flag anything you assumed."*
- **Small steps**: one change, run, read, correct. Never accept a 300-line dump blind.

Vague in → generic out

❌ *"Analyse my sleep data."*

✅ *"sleep.csv has bedtime and wake time in Geneva local time. Plan how you'd compute sleep duration per night and plot the nightly mean. Don't write code until I approve."*

ORGANISE THE WORK

Work in modules: 1 task = 1 agent

- **Cut the project into small tasks**, each with a clear goal and a check: *"load and clean"*, *"sleep per night"*, *"figure 2"*.
- **One agent per task**: a fresh session, its own branch. Small context, fewer confusions; a failed task is thrown away alone.
- **You coordinate**, like a PI with capable students: brief, review, merge (Part 3).



THE CONTROLS

Three dials, whichever agent you use

DIAL	SETTINGS	USE WHEN
1 · Edit mode: how much rope	**Ask** (touches nothing) · **Plan first** (writes a plan, waits) · **Edit** (you accept each change) · **Auto** (edits and runs on its own)	Exploring → ask · anything big → plan · results that matter → edit · throwaway attempts → auto, **only because git makes it reversible**
2 · Model: the right brain	Frontier / "thinking" models · fast, small models	Statistics, study design, subtle bugs → frontier · boilerplate → fast · stuck → escalate
3 · Steering: hand on the wheel	Interrupt · correct with specifics · paste the error · clear a stale thread	The moment it drifts: a drifting agent rarely fixes itself. Stop it, re-aim, continue

At the hack: your own agent (Claude Code, Codex, GitHub Copilot, Cline), or Cline on open models served on the HPE / NVIDIA environment.

A WORKED RHYTHM

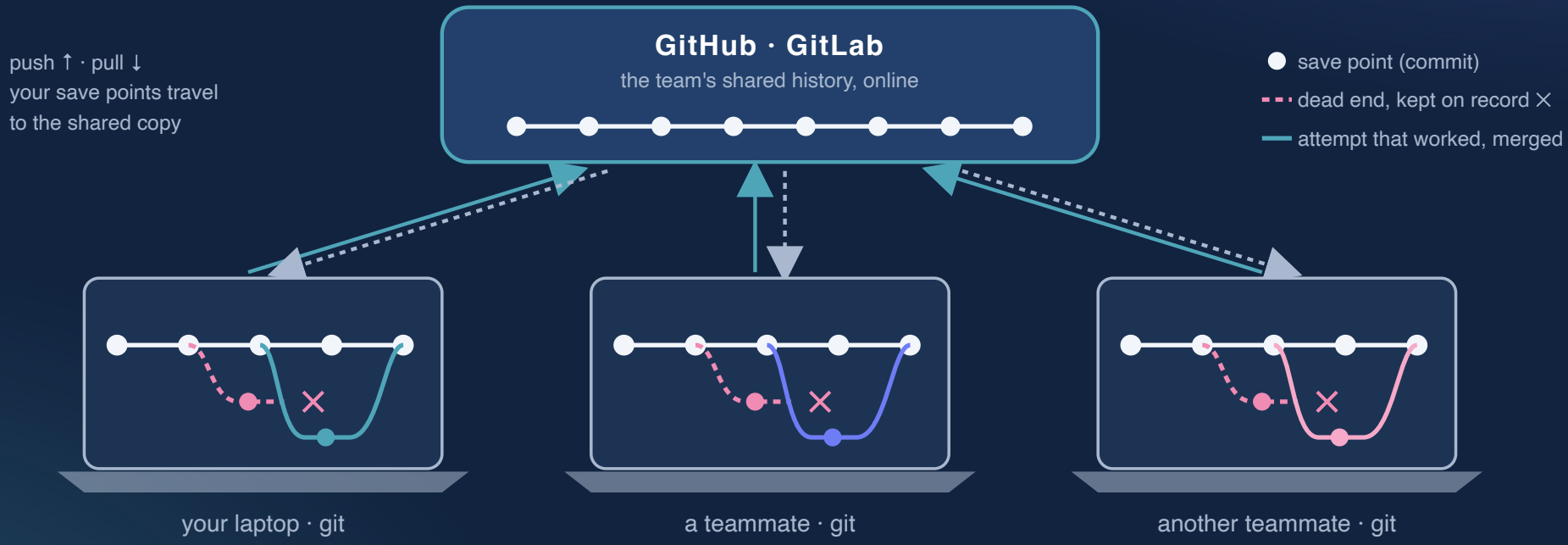
Right mode, right model, hand on the wheel

1. **Plan mode + a strong model** → *"plan how you'd test this; don't write code yet."*
2. Approve the plan → **save a starting point** (Part 3: the agent does it for you).
3. **Edit mode, small steps** → accept good changes, ask *"what changed?"* after each.
4. Stuck or slow? **Switch model** or **interrupt and re-aim**.
5. Mechanical follow-ups (tests, formatting, tidying) → **auto mode**, on a separate branch.

> That's directing.

Git & GitHub, without the jargon

Git is version control: every version of your project, dead ends included: your lab book, written by the agent.
GitHub or GitLab is where most of it happens: online, shared with your team.



Git is the lab book of the agentic era

- **Directing is exploring**: try an approach, let it fail, try another. Dead ends are part of the work.
- **Every attempt is an entry**: what was tried, why, what came out, dated.
- **Go back in one sentence**: *"the Tuesday version was better: return to it"*, without losing what you learned since.

GIT = VERSION CONTROL

The software that records **every version** of your project folder, as dated save points with a note. It runs on your laptop. **The lab book itself.**

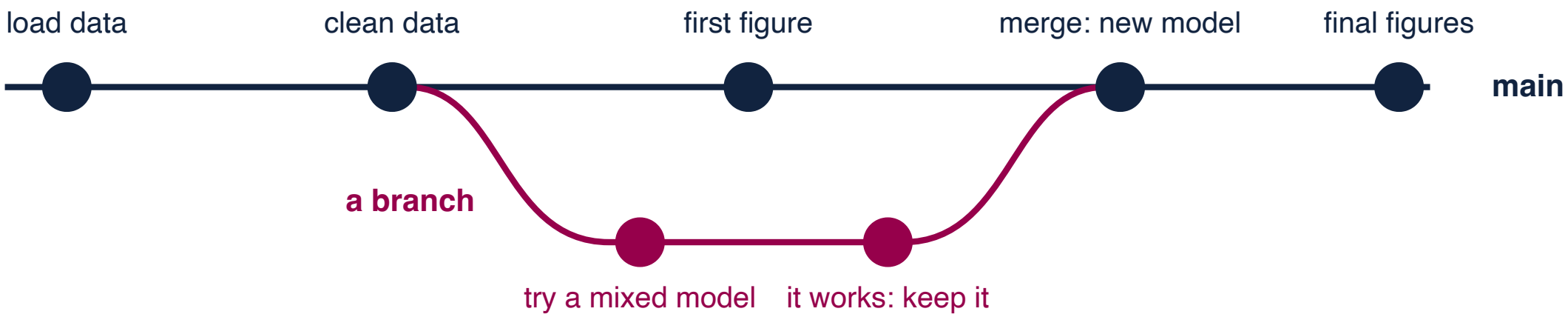
GITHUB · GITLAB = WHERE MOST OF IT HAPPENS

The websites that keep your project's history **online**: shared with your team, backed up, reviewed. GitLab is often the university's own server. At the hack, every team works in a GitHub repository.

Git is your lab book; GitHub or GitLab is the shelf where the team's lab books live.

THE WHOLE IDEA IN ONE PICTURE

Git is a time machine for a project folder



● Commit = a save point

A snapshot of the whole folder, with a one-line note: *"clean data: removed non-wear days"*.

Branch = a parallel line

Try something risky without touching the version that works.

Merge = bring it back

If the attempt works, fold it into **main**; if not, drop the branch.

THE DICTIONARY · TEN WORDS YOU WILL HEAR

Git and GitHub words, in plain English

WORD	PLAIN MEANING	EVERYDAY ANALOGY
Repository ("repo")	A project folder that remembers its whole history	A lab notebook that never loses a page
Commit	A save point, with a note saying what changed	A dated, signed notebook entry
Diff	What changed since the last save point (removed in red, added in green)	Track changes in Word
Branch · **main**	A parallel line of work · the line that works	A draft copy · the published version
Merge	Fold a branch's changes back into main	Accepting tracked changes
GitHub	A website that hosts a copy of your repo: backup, sharing, review	Google Drive, for projects with history
Push · **pull** · **clone**	Send your save points to GitHub · get others' · download a whole repo	Upload · sync · download
Pull request ("PR")	A page asking "please review these changes and merge them into main"	Sending a draft to your supervisor
`.gitignore`	A list of files git must never save or share (data, passwords)	A "do not photocopy" stamp

TWO DIFFERENT JOBS

Git lives on your laptop; GitHub is the shared copy

Your laptop

The project folder with its save points. Works offline, private, no account needed. **Protects you from yourself.**

push →
← pull

GitHub

The online copy: survives a dead laptop; one place for the team; pull requests for review. **Protects you from the world.**

← pull
push →

Your teammates

Each has their own copy and their own branches; GitHub keeps everyone in sync.

PRIVATE REPOSITORY	PUBLIC REPOSITORY
Only you and invited collaborators	Anyone can read and copy it
Start here while the work is live	At publication, for reproducibility, after checking the history

⚠️ Public is permanent: a password or a dataset pushed in public may be copied before you delete it. Data stays out of git either way.

THE LOOP THAT MAKES IT SAFE

The vibe-coding loop *is* the git loop

1. **The agent changes something** (you asked it to).
2. **You ask: "what changed?"** It summarises in plain words; the detailed diff is there if the result matters. *This is your System 2 checkpoint.*
3. **Good? "Save it."** A save point with a clear note.
4. **Bad? "Undo that."** Back to the last save point, nothing lost but the bad change.

> **Save before every big request**: a guaranteed way back.

YOU SAY	THE AGENT RUNS
"What did you change?"	`git status` · `git diff`
"Save it: cleaned the units"	`git add` · `git commit -m "…"`
"Undo that"	`git restore .`
"Show me the history"	`git log --oneline`

You don't need to remember the right-hand column. You do need to know it exists, and to read what the agent tells you it did.

Try the risky thing on a branch

- Want the agent to attempt a big refactor, or a second way of analysing the data? Ask for it **on a branch**.
- **main** stays known-good while the agent experiments.
- Bad attempt? **Drop the branch**: zero trace, zero regret. Good? **Merge it.**
- At the hack, each person works on their own branch: `firstname/topic`.

> Branches turn *"I hope this doesn't break everything"* into *"let's just try it."*

YOU SAY	THE AGENT RUNS
"Try a mixed model on a separate branch"	`git switch -c thomas/mixed-model`
"Go back to the main version"	`git switch main`
"It works, keep it"	`git merge thomas/mixed-model`
"Forget it"	`git branch -D thomas/mixed-model`

LET THE AGENT DRIVE

Hand off 95% of git: say what you want, in plain words

YOU SAY	WHAT THE AGENT DOES FOR YOU
"Start a branch for this task."	Creates `firstname/topic` and explains it in one sentence
"Save my work with a clear message."	Commits, with a note saying what changed and how it was checked
"What changed since this morning?"	Summarises the save points and changes in plain language
"Undo what you just did."	Goes back to the last save point, after telling you what will be lost
"Get my teammates' latest work."	Pulls from GitHub; if two people changed the same lines, explains the conflict and asks which version to keep
"Send it to GitHub and open a pull request."	Pushes the branch, opens the PR, writes what was verified and what was trusted

One-time setup: a GitHub account with two-factor authentication, then `gh auth login` once on your machine. From then on the agent can run `git` and `gh` (GitHub's command-line tool) on your behalf.

THE 5% THAT STAYS YOURS

The agent runs git. You keep five decisions.

When to save

Ask for a save point at each step that works, and before every big request.

Read the summary

Every change gets a plain-language summary; read the detailed diff for anything that reaches a result.

Approve the merge

Nothing reaches **main** without your OK. Never let auto mode push to main unreviewed.

Keep data out

No data files, passwords or keys in git. The agent refuses; `.gitignore` backs it up.

Own the history

Commits carry **your** name. For credit at the hack, the history is the evidence.

> The agent drafts; **you merge**.

Write the rules once; the agent follows them every time

- The agent reads `AGENTS.md` at the start of every session (Claude Code via `CLAUDE.md`).
- It tells the agent to **run git for you, explain each step in plain words, and wait for your OK** before anything irreversible.
- Every team repository at the hack ships with one.

```
# Rules for the AI agent working in this folder
You are helping researchers who may not know git.
You run git and GitHub for them; they decide.
Explain every git step in one plain-English sentence.

- Starting a task: create a branch <firstname>/<topic>.
- Saving work: after each step that works, commit with a
  plain-English message. Tell the human "Saved: …".
- "Undo that": go back to the last saved state, after saying
  what will be lost.
- "Send it to GitHub": push and open a pull request saying
  what was verified and what was trusted.
- Never force-push, rewrite history, or merge into main
  without the human's explicit OK.
- Never commit data files, passwords or keys.
```

POLL 3 OF 6 · THE LITERATURE

Soon an agent can re-run any published analysis from its repository in minutes. What happens to your field's literature?

- **A** Replication becomes routine and the literature gets cleaner
- **B** A wave of corrections and retractions, then a cleaner literature
- **C** Little changes: nobody is rewarded for checking
- **D** AI-generated papers multiply faster than anyone can check them

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20 min

18:00 · PART 4 OF 7 · OBJECTIVE 2 · UNDERSTAND

Spot the trap

Two analyses an agent ran cleanly: no error, a plausible number. Vote on what went wrong, then talk: where would the same trap hide in your data?

SPOT THE TRAP · 1 OF 2 · SLEEP

An agent computes hours slept per night for 150 students from bedtime and wake-up times (Geneva clock). Everyone seems to sleep about 1 hour less on the night of 24 → 25 October. Most likely?

- **A** Exam stress that week
- **B** The sleep app crashed that night
- **C** The clocks went back at 03:00: subtracting clock times loses an hour
- **D** Random noise

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C: no error, no warning, one hour lost

- Mean sleep **6.9 h** that night with clock times, against **7.9 h** in UTC, in line with the other Saturday nights (**7.7 h**).
- Caught by **a plot per night** and **the calendar**: domain knowledge, not code.
- Ask the agent: *"convert to UTC before subtracting, then plot one value per night."*

Discuss · 2 minutes

Where does **time** hide in your data? Device clocks, sampling rates, recordings across midnight, time zones in multi-site studies… Unmute, or write it in the chat.

SPOT THE TRAP · 2 OF 2 · MISSING ANSWERS

An agent averages every evening mood answer in the phone study: 4.6 out of 7, "students coped well with the term". But 37% of the prompts went unanswered. The true average is most likely...

A About the same: missing answers are random

B Higher: happy, busy students skip the prompt

C Lower: students skip it on bad days, and go silent when struggling

D Impossible to say anything

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C: the missing answers are not missing at random

- The evening before a skipped prompt, mood was **4.3**; before an answered one, **4.8**.
- The quarter who answer least start the term with PHQ-9 **10.7**, the others **6.9**.
- 32 students go silent before the last two weeks: early mood **4.2**, against **5.0**.
- Ask the agent: *"who is missing, and are they different? Compare what we know about them."*

Discuss · 2 minutes

Where does **missing data** hide in your field? Patients who miss visits when sicker, trial dropouts, wearables taken off on bad days, scans excluded for movement… Unmute, or write it in the chat.

Both agents were **mostly right**: the dangerous kind of wrong. What catches them is a skeptic asking one more question.

FROM THE COURSE HANDOUT · KEEP THESE OPEN

Five prompts worth stealing

1. *"Explain what this code does before you change anything."*
2. *"Outline your plan, then wait for my OK."*
3. *"Write a test that would fail if this result were wrong."*
4. *"What could make this conclusion wrong? Argue against yourself."*
5. *"Summarise in plain language what you changed and what you assumed."*

> And for git, just say it: *"start a branch"* · *"save my work"* · *"what changed?"* · *"undo that"* · *"send it to GitHub"*.

POLL 4 OF 6 · TRAINING

A first-year PhD student can now get working analysis code from an agent in minutes. In their first year, they should…

A Use agents freely: directing them is the skill that matters now

B Write their core analyses by hand first, agents later

C Use agents, but explain every line at lab meeting

D Keep agents away from thesis-critical analyses

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8 min

18:20 · PART 5 OF 7 · OBJECTIVE 2 · UNDERSTAND

Review hunt

An agent wrote an analysis that runs cleanly. Would you sign it? Find what is wrong, together.

An agent wrote this. Would you sign it?

1. Open `review-hunt/analysis.py` and the codebook `phone-study/data/README.md`.
2. Read the output on the right (or run it).
3. Hunt: the code, the data, the method, the summary.
4. In the chat: every problem, and **what tipped you off**.

The prompt the agent got is at the top of the file. Allowed: plots, the codebook, a second AI, your judgment.

```
Analysed 7314 participant-days from 150 participants.

1. Associations with next-day mood (Pearson r)
   sleep duration : r = +0.10
   daily steps    : r = +0.14

2. Low-mood day classifier (random forest,
   30% held-out test set)
   AUC = 0.73

Summary
   Physical activity is the stronger correlate
   of mood (r = +0.14 vs +0.10 for sleep):
   students who walk 10,000 steps a day have
   half as many low-mood days.
   Phone data flag low-mood days with high
   accuracy (AUC = 0.73), supporting passive
   sensing as an early-warning signal for
   depressive episodes.
```

THE ANSWER KEY

Five problems, and the headline flips

#	PROBLEM	EFFECT	WHAT CATCHES IT
1	Each night paired with the **next** evening's mood	r(sleep, mood) +0.35 → +0.10	the codebook: the night ends on the row's morning
2	Phone left lying kept as 0 steps	zeros fake a steps–mood link: r +0.09 → +0.15	a histogram: spike at 0
3	Same students in train and test	AUC 0.81 → **0.63** on new students	"would it work on a new patient?"
4	A number no line computed	"10,000 steps, half as many low-mood days"	trace every number to the output
5	Overclaim	a low-mood *day* is no depressive *episode*; PHQ-9 sits unused	what does the outcome measure?

Fixed: sleep goes with mood (r ≈ +0.35), steps add little (r ≈ +0.09), the classifier generalises modestly to new students (AUC ≈ 0.63). Same phone study as the exercise and both traps: one codebook, many ways to go wrong.

WORKING TOGETHER IN A LAB

When code is cheap, review is the work

Peer review matters more

No AI-generated analysis reaches a paper without a second human reading the reasoning and the checks.

Show how you checked

Short sprints, visible demos: what you built *and how you verified it*. Pair on the prompts.

Share the rules, track provenance

A lab `AGENTS.md`, shared environments; whose prompt, which model, which data version.

> The failure mode is **unreviewed AI code flowing straight into results**.

POLL 5 OF 6 · THE SCARCE SKILL

In 2031, the scarcest skill in a neuroscience lab will be...

- **A** Programming
- **B** Statistics and methods
- **C** Choosing the question worth asking
- **D** Judging whether a result is true
- **E** Collecting good data from real people

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10 min

18:28 · PART 6 OF 7 · OBJECTIVE 3 · PREPARE

The hack, concretely

How the 32 hours run, how each challenge is set up, and what our compute partner brings.

FRIDAY 6 → SATURDAY 7 NOVEMBER · CAMPUS BIOTECH

32 hours, hour by hour

FRI 12:30

Opening

Coffee, no lunch. Challenges presented at 13:00; your team is already formed.

FRI AFTERNOON

Build

Explore, try, fail, keep what works: tonight's method.

BEFORE DINNER

Cross-pitch

Each team shows its progress to the others; steal good ideas.

NIGHT

Optional

Sleep on site if you said yes: bring a mat and a sleeping bag.

SAT ≈ 11:30

Peer review

Teams review each other in rotation, then tag the `demo`.

SAT 17:00

Public finals

At Campus Biotech, 17:00 → 18:30, with the Dean: your pitch and your demo.

Before 6 November: vote and confirm your place (23 → 28 Oct) · a GitHub account with two-factor authentication · Crash Course #3 on grant writing, Tue 3 Nov, 17:00 → 19:00.

HOW IT WORKS

One GitHub repository per challenge

READY BEFORE YOU ARRIVE

Data and methods

We do our best to prepare each challenge in advance, with its runner: the dataset (or a synthetic stand-in), the methods to start from, and the questions worth asking.

EVERYTHING DOCUMENTED

In the repo, or online

The challenge repository holds the README, the data description and an `AGENTS.md` your agent reads. The rest is on **facmed-hack.sparkboard.com**.

HOW THE TEAM WORKS IN IT

Branches and pull requests

One branch per task (1 task = 1 agent), merged through pull requests, as in Part 3. Real data never goes into git.

> Each challenge has a **lead** (a PI) and a **runner** (a PhD student or postdoc) who knows the data and the repository.

What HPE brings to the hack

Resources

- **8 NVIDIA GPUs** for the weekend, and compute after it for the projects that continue.
- **Open AI models served on them**: a coding model for your agent, and a model for audio and video.
- **A ready-made VS Code in the browser**, with an agent already set up: nothing to install.
- **Your challenge's data**, stored there and read-only; heavy analyses run as jobs on the GPUs.

Expertise

- **Three HPE engineers on site**, all weekend: compute, models, getting your job to run.
- **Privacy layers** for sensitive inputs, and advice on keeping data where it belongs.

Only public, synthetic or anonymised data on the HPE environment during the hack. UNIGE participants also get weekend access to the university's HPC cluster.

Record the brainstorm; let AI write the next steps

1. **Record** the conversation on your phone, with everyone's OK.
2. **Transcribe** it with AI.
3. **Synthesise**: ask an agent for the decisions, the tasks (who, by when) and the open questions.

> **Stay pointed at your final goal.** Say it out loud when you start recording; give it to the AI with the transcript and ask what drifted away from it.



No patient information in a recording that goes to a cloud AI: same rule as the data.

22 min

18:38 · PART 7 OF 7 · OBJECTIVE 1 · IMAGINE

AI governance in science: let's keep talking

Nobody has settled the rules yet. Labs are writing them, fields are arguing about them. Tonight, three breakout rooms write some of them.

Governance starts in the lab

Learning first

Do by hand what builds your core skills; hand off what you don't need to learn.

Draft before AI

"You must always draft a text yourself before handing it to AI."

Verify by falsifying

"Know how you can falsify what a model produces."

Limit the risks

"Participant data should not be shared with AI tools."

> *"Our lab policy's biggest effect wasn't any single rule, but rather creating a culture in which we discuss AI use rather than hide it."* Nicolas Schuck

FROM THE LAB TO THE FIELD

Five rules for each of us, norms for all of us

1. **Confidentiality**: no patient or unpublished data in uncleared tools.
2. **Provenance**: commit the code, keep the prompts.
3. **Correctness is yours**: verify anything that reaches a result.
4. **Disclosure**: say how AI was used; it is never an author.
5. **Keep your skills**: enough to review what it writes.

> *"We should not only have lab policies but decide as a field where we stand on the shifts in money, priorities and agenda that come with AI."*
>
> Nicolas Schuck

POLL 6 OF 6 · DEPTH

Pre-read 1 warns that AI coding could make theoretical neuroscience *"prolific but shallow"*. For your own field, you think…

- **A** It is already happening
- **B** It will happen unless incentives change
- **C** The opposite: faster coding frees time to think
- **D** Too early to say

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BREAKOUT · 10 MINUTES · THREE ROOMS, YOUR CHOICE

Three rooms, three questions

ROOM 1 · IN THE LAB

Write your lab's AI policy

What do trainees still do by hand? What must be checked, and by whom, before an agent's result counts?

ROOM 2 · IN THE FIELD

Set the norms

What should journals, funders and reviewers ask for: disclosure, credit, re-running the code?

ROOM 3 · WITH PATIENTS' DATA

Draw the lines

Which data may never meet an AI tool? Who decides, and what do participants consent to?

Pick a rapporteur in the first minute. Come back with **two concrete proposals**, for the main-room chat.

DIRECTOR, NOT AUTHOR

Let's keep talking

What becomes possible

More time for the questions and for patients. Science for more people. A literature anyone can re-run.

What it takes

Judgment kept sharp. Norms set openly, in labs and as a field. Sensitive data kept where it belongs.

Where we continue

At the hack, 6–7 November. In your lab: write your AI policy together. The course: github.com/wazaahhh/vibe-coding-crash-course

> **Report back**: two proposals per room, from lab, field and data. Then, in the chat, one last line: *"When agents write the code, the part of my research that stays mine is …"*

A LAST WORD

"AI is obviously going to have a massive impact on how we do science, and it's hard to know what the long run looks like."

Anthony Zador · Cold Spring Harbor Laboratory · *The Transmitter*, June 2026

So keep the conversation going: in your lab, and on 6 November.